Representing Inequalities as Regions

Exam code: 0607

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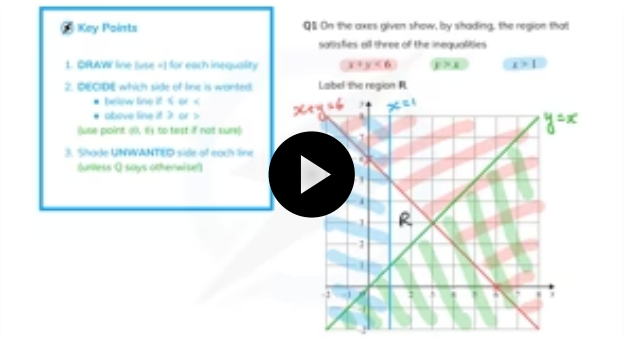

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Finding regions using inequalities

What are 2D inequalities?

- A **2D inequality** involves two variables
  - E.g. $y < x$ or $x + y \geq 8$
- The solution to a 2D inequality is a **region** in the xy plane
- The solution to a **1D inequality** can also be represented as a region in the xy plane
  - E.g. $y \geq 2$ represents the region of points that lie on or above the line $y = 2$

How do we draw inequalities on a graph?

- A set of **2D inequalities** can be shown **graphically** using **straight lines** and **shaded regions**
- To draw the correct lines:
  - Replace the inequality sign with "=" and draw that line
    - Use a **solid** line for ≤ or ≥ (to indicate the line is included)
    - Use **dotted** line for < or > (to indicate the line is not included)

Help me understand
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- if "$y \geq$ ..." or "$y >$ ..." then the wanted region is **above** the line
- If you are unsure
  - substitute the coordinates from a point on one side of the line into the inequality
  - determine whether or not the inequality holds true on that side

- For **vertical lines**:
  - the wanted region for $x < k$ is to the **left** of $x = k$
  - the wanted region for $x > k$ is to the **right** of $x = k$

- To do the shading:
  - **Shade the unwanted sides** of each line (unless the question says otherwise)
    - You are shading away any parts you don't want
    - This will leave behind a clear region that is the wanted region (rather than trying to look for the wanted region under multiple shades)
    - Label the wanted region *R* (unless the question says otherwise)

Worked Example

Show, graphically, the region that is satisfied by all three inequalities below:

$$3x + 2y \geq 12 \quad y < 2x \quad x < 3$$

Label this region *R*.

Answer:

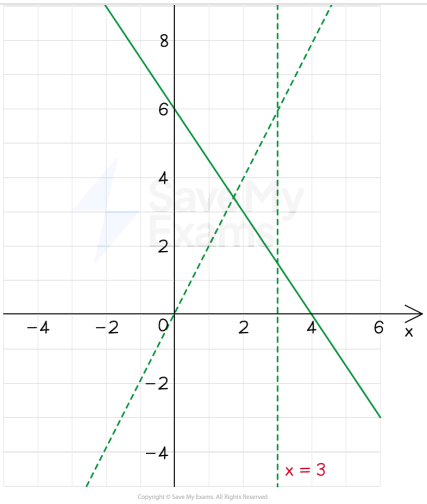
First draw the three straight lines: $3x + 2y = 12$, $y = 2x$ and $x = 3$

Use your knowledge of **Straight Line Graphs**, $y = mx + c$
You may wish to rearrange $3x + 2y = 12$ to the form $y = mx + c$ first

$$2y = -3x + 12$$
$$y = -\frac{3}{2}x + 6$$

The line $3x + 2y \geq 12$ is a solid line because of the "≥"

The lines $y < 2x$ and $x < 3$ are dotted lines because of the "<"



Now we need to shade the **unwanted** regions

For $3x + 2y \geq 12$ (or $y \geq -\frac{3}{2}x + 6$), the unwanted region is below the line
We can check this with the point (0, 0)

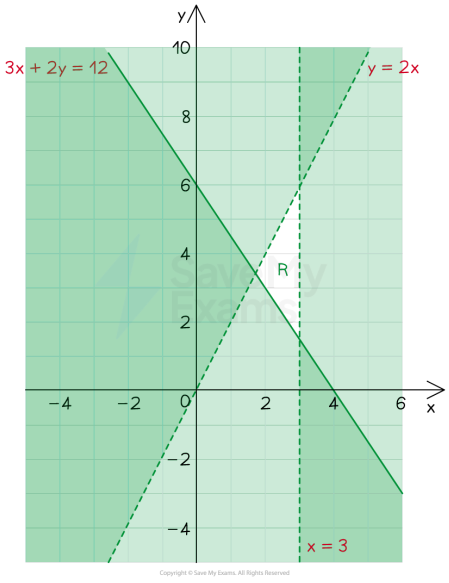
"$3(0) + 2(0) \geq 12$" is false therefore (0, 0) does lie in the unwanted region for $3x + 2y \geq 12$

For $y < 2x$, the unwanted region is above the line
Check with another point, for example (1, 0)

"$0 < 2(1)$" is true, so (1, 0) lies in the wanted (i.e. unshaded) region for $y < 2x$

For $x < 3$, shade the unwanted region to the right of $x = 3$
If unsure, check with a point

Finally, don't forget to label the region *R*



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